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# Is Mergeability Algorithm-Relative?

## Predicting Model Merging Success from Weight- and Representation-Space Signals

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### Abstract

Model merging combines models fine-tuned from a common checkpoint by arithmetic on their weights, but whether a merge works is usually discovered only after performing it. We ask whether the pre-merge signal that best predicts merging damage depends on the merging algorithm. We fine-tune 70 models on CIFAR-100 task families whose similarity is set by a five-level dial, run 3,864 merges with weight averaging, task arithmetic, TIES and DARE, and test five signals against trivial baselines. The answer is no: the same similarity signals predict the damage for all four algorithms. CKA is the most reliable diagnostic and the only one that keeps its sign on a ResNet-18 control, where weight-space cosine reverses. We also show that weighted sign conflict is a function of cosine, and that DARE’s apparent failure is an untuned scaling.

### 1. Introduction

Fine-tuning a pretrained model  $\theta_0$  on task  $i$  gives  $\theta_i = \theta_0 + \tau_i$ , where the *task vector*  $\tau_i$  records what the task changed. Model merging builds one multi-task model by combining task vectors directly, with no data and no training (Wortsman et al., 2022; Ilharco et al., 2023). It often works but fails silently: one finds out only by evaluating the merged model. Since merging algorithms work in different ways, they may fail for different reasons, which raises our question: *does the signal that predicts merging damage depend on the algorithm?* If yes, the diagnostic must match the merger; if one signal dominates everywhere, mergeability is a property of the models and one diagnostic suffices; if nothing beats trivial baselines, the diagnostics themselves are in doubt. We wrote down in advance which signal we expected to matter for each algorithm (Table 1), built task families whose similarity we control by design, and tested the signals against

baselines, with a stability control across seeds and architectures. Code and results: <https://github.com/vulpiani1948744-lab/DLAI-final.git>.

### 2. Related work

Model soups average fine-tuned weights (Wortsman et al., 2022); task arithmetic adds scaled task vectors (Ilharco et al., 2023); TIES trims small updates and resolves sign disagreements, which it identifies as a cause of interference (Yadav et al., 2023); DARE drops most delta parameters at random and rescales the rest (Yu et al., 2024). Linear CKA compares the representations of two networks (Kornblith et al., 2019). We turn these explanations of interference into predictors and test them across algorithms.

### 3. Method

**A dial for task similarity.** CIFAR-100 (Krizhevsky, 2009) is split into 5 tasks of 20 classes (10,000 training and 2,000 test images each). In the *semantic* split each task joins four related superclasses (e.g. aquatic life, or outdoor scenes and vehicles), so the tasks need different features and merging should be hard; in the *random* split every task is a miniature of the dataset and merging should be easy. A mixing fraction  $\alpha \in \{0, .25, .5, .75, 1\}$  reassigns a random  $\alpha$  share of the classes, keeping 20 classes per task; semantic purity falls 100/80/62/41/33%. Images, budget, hyperparameters and seeds (0, 1) are identical across levels.

**Models and frozen heads.** We use ViT-Tiny (Dosovitskiy et al., 2021) (ImageNet-pretrained, 5.5M parameters) at  $128 \times 128$  (linear probe 71.5% vs 78.7% at 224, for a third of the tokens). For each task we fit a logistic-regression head on frozen  $\theta_0$  features and *freeze* it before fine-tuning the encoder (5 epochs, AdamW, learning rate  $10^{-4}$ , cosine schedule): since every encoder is read out by heads matched to  $\theta_0$ , a post-merge drop measures merging damage, not a mismatched head. Fine-tuning gains 8.2–19.9 points over the probe. As a control we repeat this with ResNet-18 (He et al., 2016) on the two extreme levels, with

BatchNorm statistics frozen and excluded from  $\tau$  (left free, they corrupted the task vectors). This gives 50 ViT and 20 ResNet task vectors, trained on the same GPU and checked for anomalous norms.

**Merging algorithms.** The merged encoder is  $\theta_0 + \tau_S$  for a subset  $S$  of tasks. *Averaging*:  $\tau_S$  is the mean of the  $\tau_i$ . *Task arithmetic*:  $\tau_S = \lambda \sum_i \tau_i$ ,  $\lambda \in \{.1, .2, .3, .5, 1\}$ . *TIES*: keep each  $\tau_i$ 's top 20% entries, elect one sign per parameter, average the agreeing entries;  $\lambda \in \{.5, 1\}$ . *DARE*: keep a random 10% of each  $\tau_i$ , rescaled by 10, then TIES;  $\lambda \in \{.2, .3, .5, 1\}$ , the values below 1 added after a first run (Sec. 4). Averaging is task arithmetic with  $\lambda = 1/|S|$ .

Table 1. Pre-registered predictions: the signal expected to matter for each algorithm, its  $\rho$  on ViT, and the verdict (Sec. 4).

| algorithm   | predicted signal | $\rho$ | verdict             |
|-------------|------------------|--------|---------------------|
| Averaging   | drift            | -0.01  | failed              |
| Task arith. | cosine           | 0.74   | holds, not specific |
| TIES        | sign conflict    | -0.68  | same as cosine      |
| DARE        | effective rank   | -      | not computed        |

**Signals and baselines.** We test five signals, all computed before merging from the subset's task vectors (averaged over its pairs) and 2,000 unlabelled test images: the *cosine* between task vectors; the *sign conflict*, the share of  $\sum_k |\tau_{a,k} \tau_{b,k}|$  on coordinates of opposite sign; the *subspace overlap* of the top-8 singular directions of each layer's update; the *CKA* between the features of two fine-tuned encoders; and the *drift*, the CKA of each encoder with  $\theta_0$  (high = little drift). A signal is useful only if it beats three trivial baselines: always predicting the mean (b0), the number of merged tasks (b1), and the task-vector norm  $\|\tau\|$  (b2).

**Outcome and statistics.** We merge all pairs, triples and the five-task set per level and seed (2,940 ViT and 924 ResNet merges). The outcome is *normalized accuracy*, the mean over merged tasks of merged over fine-tuned test accuracy, at each algorithm's best setting (a fixed setting changes no correlation by more than 0.02). Spearman  $\rho$  is computed within pairs and within triples and averaged, which removes b1; partial  $\rho$  given  $\|\tau\|$  addresses b2. Stability is the Kendall  $\tau$  between signal rankings; generalization is tested by ridge regression fitted on four similarity levels and tested on the fifth.

## 4. Results

**The dial works.** Merging gets easier as tasks become alike (Fig. 1a): five-task merges rise from 92.8% to 96.4%, pairs from 96.7% to 98.2%. No merge fails outright at its best setting (ViT minimum 92.4%), so the signals must

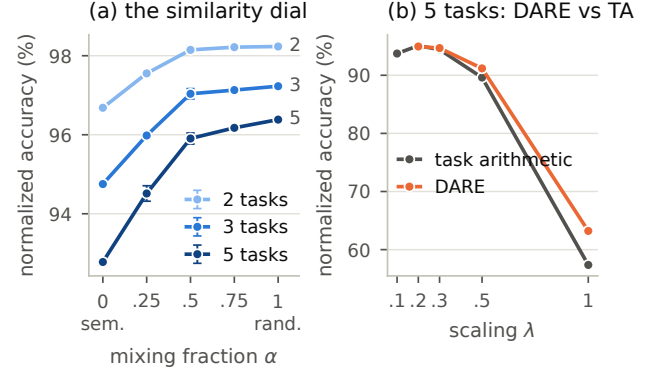


Figure 1. ViT-Tiny. (a) Normalized accuracy against the mixing fraction (mean of the four algorithms at their best setting; bars: two seeds). (b) Five-task merges against  $\lambda$ .

rank differences of a few points.

**The same signals for every algorithm.** On ViT, cosine, sign conflict, subspace overlap and CKA predict all four algorithms alike,  $|\rho| = 0.68$ –0.78 (Fig. 2), with differences inside the bootstrap intervals (e.g. CKA with task arithmetic,  $[0.64, 0.79]$ ); drift predicts nothing ( $|\rho| \leq 0.03$ ). The ranking among the four is not reproducible across seeds (Kendall  $\tau$  from -0.20 to 0.20) because they are nearly collinear ( $|\rho| \geq 0.93$  between any two): which one wins for an algorithm is noise. On ResNet-18 the ranking is stable across seeds ( $\tau = 0.74$ –0.95) and the same for every algorithm ( $\tau \geq 0.80$ ).

**Beyond the baselines.** The norm alone predicts damage ( $\rho = -0.56$  to  $-0.49$ ), but the similarity signals survive conditioning on it (partial  $|\rho| = 0.55$ –0.69) and predict an unseen similarity level: subset size and norm reach  $R^2 = 0.41$ –0.44, adding CKA 0.65–0.69, the best single signal. Within one level, however,  $|\rho|$  falls to 0.20–0.31.

**Across architectures.** On ResNet-18 cosine *reverses* sign ( $\rho = -0.79$  to  $-0.72$ ) and vanishes given the norm (partial  $\rho = -0.11$  to 0.04): there it only measures distance travelled. Drift, silent on ViT, is the strongest signal ( $\rho = 0.90$ –0.92). CKA is the only signal positive and robust to the norm on both backbones (partial  $\rho = 0.58$ –0.68 on ViT, 0.55–0.68 on ResNet).

**Two findings about the methods.** With  $S_{\pm}$  the sums of  $|\tau_{a,k} \tau_{b,k}|$  over coordinates of equal (+) and opposite (-) sign, sign conflict is  $c = S_{-}/(S_{+} + S_{-})$  and

$$\cos(\tau_a, \tau_b) = (1 - 2c) \cos(|\tau_a|, |\tau_b|),$$

so the two signals agree in rank (-1.00 here): the predictions for TIES and task arithmetic cannot be separated.

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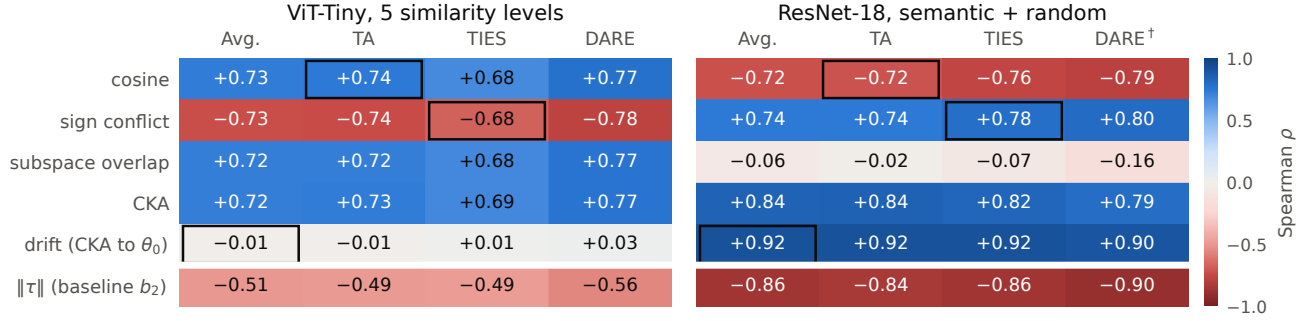


Figure 2. Spearman  $\rho$  of each signal with normalized accuracy, within subset size; boxes mark the pre-registered signals. <sup>†</sup>DARE at  $\lambda = 1$  only.

Second, with 90% of the entries dropped DARE’s surviving supports barely overlap, so TIES’s disjoint mean turns into a *sum*: DARE tracks task arithmetic at the same  $\lambda$  (Fig. 1b, within 1.6 points for  $\lambda \leq .5$ ), collapses at  $\lambda = 1$  (63.2% on five tasks) and reaches 95.0% at  $\lambda = 0.2$ . Only two of our four mechanisms are truly distinct.

## 5. Discussion and conclusions

In this setting mergeability behaves as a property of the models, not of the algorithm: one family of similarity signals predicts the damage for every algorithm, and CKA – the best predictor of an unseen similarity level and the only one consistent across a Transformer and a CNN – is the diagnostic to use. Of the pre-registered predictions (Table 1) only the one for task arithmetic held, and not specifically. The absence of algorithm-relativity has two causes worth designing around: redundant signals, and algorithms that reduce to task arithmetic.

**Limitations.** Merges never failed badly (at least 92.4% on ViT and 89.5% on ResNet at the best setting), and most predictive power separates similarity levels rather than subsets within one. Models are small, on one dataset at  $128 \times 128$ , with two seeds; subsets share tasks, so bootstrap intervals are optimistic. The ResNet control covers two levels with DARE at  $\lambda = 1$  only, and hyperparameters were selected on the evaluation split. A heterogeneous benchmark, a layer-wise CKA ablation and effective rank were planned but not done. The DARE sweep was recomputed on another device from the same task vectors and matched the original run.

## 6. Statement on the use of AI

Claude (Anthropic) served as a coding and writing assistant for this project. It supported the refinement of the experimental methodology, assisted in extending and debugging

the codebase, and aided in source verification and bibliographic assembly. It was also used to format and summarize drafts in the final report. References were checked against the official proceedings, every value in the text is computed from the result files, and predictions that failed are reported unchanged. We have reviewed all methods and results and are able to explain them.

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